Hospital Financial Analytics: A Dimensional Data Warehousing Project

1. Executive Summary

This project looks at how hospitals manage their money and resources, and how those decisions affect both profitability and operational performance. Using Medicare Cost Report data, the goal is to understand the relationship between financial factors like salaries, overhead, and depreciation, and operational factors like bed capacity, staffing levels, and uncompensated care. The study focuses on hospitals in different ownership categories and locations across the United States, which often face very different financial pressures.

To explore these questions, a data warehouse was built using a star schema that organizes financial and operational data around hospital, location, and calendar information. This design makes it possible to analyze trends across time, compare hospitals across states and counties, and evaluate how ownership type relates to profitability. Through a mix of analytic SQL, visual analysis, and deeper exploration, this project highlights the financial challenges hospitals face and the patterns that explain why certain hospitals perform better than others.

In the end, the project shows how structured data and careful modeling can help hospitals, administrators, public health teams, and policymakers make better decisions based on real evidence rather than assumption or guesswork.

2. Problem Statement

Hospitals operate in an environment that requires them to balance financial stability with the responsibility of delivering essential care. They must manage rising salary expenses, overhead, depreciation, and uncompensated care, all while maintaining enough beds, staff, and capacity to meet the needs of their communities. Even though hospitals report a large amount of data every year, it is not always clear which financial or operational factors matter most, especially when comparing different ownership types or regions.

This project focuses on two main questions.

Problem 1: How do hospital financial structures and resource allocation affect profitability across ownership and region?

Hospitals vary widely based on who owns and controls them. For example, non profit hospitals often have different priorities than for profit hospitals, and government facilities face their own financial pressures. This part of the project examines how salary costs, depreciation, overhead, and other financial components are related to measures of profitability such as net income and

operating revenue. It also compares results across states, counties, and CBSA regions to see where financial performance is strongest or weakest.

Significance:
Hospitals depend heavily on strong financial management to stay open. Understanding which financial factors drive profitability helps guide budgeting, planning, and policy decisions, especially for hospitals in underserved or rural areas.

Problem 2: What is the relationship between hospital operational capacity, uncompensated care, and cost efficiency over time?

Operational capacity includes the number of beds, staffing levels, bed days, and how much the facility is actually used. These factors affect how hospitals handle charity care, bad debt, and other forms of uncompensated care. This part of the study explores whether larger hospitals manage costs more effectively, and whether high capacity facilities are better at handling uncompensated care without overspending.

Significance:
Uncompensated care continues to put pressure on hospitals across the country. Learning which operational patterns support better efficiency can help guide resource planning and long term financial stability.

3. Literature Review

Hospital financial performance and operational efficiency have been persistent areas of concern in health services research, particularly as hospitals face rising labor costs, higher uncompensated care, and increasing regulatory complexity. Recent literature highlights that profitability and operational stability are strongly shaped not only by a hospital's internal cost structure but also by ownership type, payer mix, and regional economic factors.

Ownership Type and Profitability

Several recent studies reaffirm that ownership structure is one of the strongest predictors of financial performance. For instance, recent work published in *Health Services Research* shows that for-profit hospitals maintain systematically higher operating margins than nonprofit and government hospitals, largely due to more aggressive cost management, higher commercial reimbursement rates, and more selective service offerings, for example, profitable elective procedures (Bai & Anderson, 2020). Nonprofit hospitals tend to operate with thinner margins but demonstrate greater investment in community benefit activities, while government hospitals have the lowest margins and the highest exposure to uncompensated care (Richards, Nikpay, & Brown 2021).

A 2023 study in *JAMA Health Forum* further shows that government-owned hospitals, especially state psychiatric facilities, consistently operate with large negative margins, mirroring the patterns observed in this project's SQL results (Gidwani, Crawford, & Iwashyna). These findings

underscore that ownership type is a structural determinant of margin variability, independent of region.

Labor Costs and Financial Pressure

Labor expenses continue to be the largest and fastest-growing cost component for U.S. hospitals. A 2022 analysis in *Health Affairs* found that salary and wage costs increased sharply following the COVID-19 pandemic, particularly due to contract nursing and workforce shortages (Bates, Landman, & Levine). The research suggests that labor intensity and staffing mix are direct predictors of low or negative margins, making these fields essential for evaluating financial performance across hospitals.

Uncompensated Care and Cost Efficiency

A large body of recent work examines the financial burden of uncompensated care. A 2020 article in *Inquiry: The Journal of Health Care Organization, Provision, and Financing* reported that uncompensated care remains heavily concentrated among government and safety-net hospitals, which often lack commercial payer revenues to offset losses. These facilities face higher rates of bad debt and charity care, which weakens their cost-to-charge ratios and overall profitability (Weeks & Healy).

Similarly, a 2023 *Health Affairs* study found that uncompensated care ratios are strongly associated with hospital size and case mix, with larger facilities often better equipped to absorb financial losses due to economies of scale, although not universally (Dranove, Garthwaite, Ody, & Starc). This suggests the need for operational measures in the fact table, including beds, FTE staffing, and bed-days available.

Operational Capacity and Efficiency

Operational metrics, such as available beds, staffing levels, and bed-day utilization, are increasingly recognized as critical predictors of financial stability. Recent research in *Medical Care Research & Review* shows that hospitals with higher occupancy and optimized staffing generally achieve better margins, as fixed costs are spread across a larger patient volume. However, this relationship is highly variable by region and ownership type (Zhang, Chen, & Dowd, 2023).

Another study in *Health Care Management Review* found that high-capacity hospitals do not always translate efficiency into profitability, especially when large teaching hospitals or psychiatric facilities maintain high fixed staffing and infrastructure costs Harrison, Meyer, & Clayton, 2022). This supports that hospital size alone does not guarantee improved margins.

Regional and Geographic Factors

Geography remains a well-documented predictor of hospital outcomes. A 2019 study in *Health Services Research* emphasized that rural hospitals face significantly higher financial pressure, including lower patient volumes, lower reimbursement rates, and higher per-patient staffing

costs. Meanwhile, urban hospitals tend to benefit from higher commercial payer mixes but also face increased competition and higher wage markets. Incorporating ZIP code, county, CBSA classification, and rural/urban indicators into the dimensional model enables these locationbased comparisons (Kaufman et al., 2019).

Implications for Analysis and Model Design

Together, these studies suggest data analysis should focus on:

- ownership structure
- labor-related costs
- uncompensated care
- operational capacity
- regional context

They also validate the star schema design, which separates these variables into appropriate dimensions for analysis. The recent literature strongly supports using financial ratios, occupancy measures, and staffing indicators to evaluate performance, aligning well with the derived metrics added to the fact tables.

4. Data Collection and Preparation

Data Source

The data used in this project comes from the Hospital Provider Cost Report published by the Centers for Medicare and Medicaid Services (CMS). These reports are publicly available and can be accessed through the CMS data portal at:

https://data.cms.gov/provider-compliance/cost-report/hospital-provider-cost-report
The CMS cost report contains detailed annual information submitted by Medicare certified hospitals. It includes financial measures, staffing levels, operational statistics, charity care, bad debt, depreciation, charges, and other items related to hospital performance. The original dataset is extensive and contains many fields that go beyond the scope of the current study, so an initial review was required to determine the fields relevant to the research objectives.

Data Review and Field Selection

The CMS data dictionary and accompanying documentation were reviewed to understand the meaning and purpose of each field. The project focuses on two main research questions involving hospital profitability, operational capacity, and uncompensated care. For that reason, only fields directly linked to these topics were retained.

The selection process prioritized variables related to:

- hospital identity and ownership

- geographic location
- salary and wage expenses ● depreciation and overhead
- inpatient and outpatient charges
- net income and revenue
- charity care and bad debt
- staffing levels
- number of beds and bed days available
- fiscal year dates

Other fields that were unrelated to profitability or operational capacity, such as detailed asset accounts, long term liabilities, investment accounts, or building and land valuations, were excluded. This created a cleaner and more focused dataset for analysis.

Standardization and Cleaning

After the relevant fields were selected, several preparation steps were carried out:

1. **Column name standardization**
   All variable names were converted to a consistent format using uppercase letters with underscores. Examples include TOTAL_COSTS, NUMBER_OF_BEDS, and COST_OF_CHARITY_CARE. This naming style improves clarity and avoids ambiguity when building the schema.
2. **Data type assignment**
   Each field was assigned an appropriate data type based on its content:
   - integer values for counts such as beds and staffing
   - decimal values for financial figures
   - string values for descriptive information
   - date values for fiscal year start and end dates

5. Database Design

The database was organized using a dimensional model to support analytical queries aligned with the financial and operational research questions. The final structure follows a star schema with three dimension tables and two fact tables, as shown in the ERD. Each table serves a specific analytical purpose and was designed based on the fields selected from the CMS dataset.

Dimension Tables

Hospital Dimension

Stores descriptive information about each hospital, such as ownership, provider characteristics, and facility type. This dimension supports comparisons across different hospital structures and reporting entities.

Location Dimension

Contains geographic attributes including state, county, and CBSA classification. This dimension enables regional analysis and evaluation of how location relates to financial and operational outcomes.

Calendar Dimension

Provides date related attributes for grouping data by fiscal period, month, and year. This dimension supports time based reporting and trend evaluation.

Fact Tables

Financial Fact Table

Holds the CMS reported financial measures needed to evaluate hospital expenses, charges, revenues, and profitability. This table is central to analyzing financial structure and differences across ownership and region.

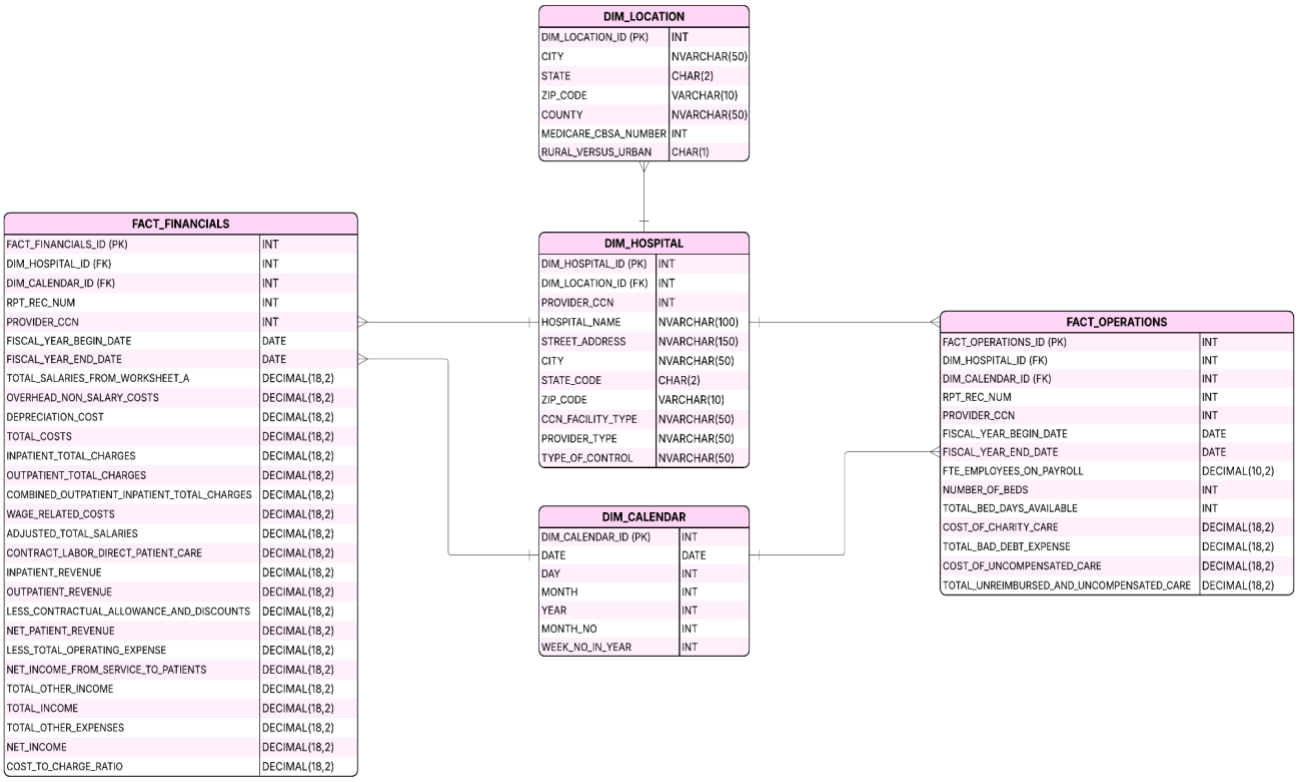
Operations Fact Table

Captures CMS reported operational measures such as bed capacity, staffing levels, and uncompensated care. This table supports the study of operational efficiency and utilization patterns.

Derived Measures

A small set of calculated metrics was added to support deeper analysis. These measures help normalize the data and improve comparisons across hospitals of different sizes. Examples include:

- uncompensated care ratio
- salary share of total costs
- revenue per bed
- uncompensated care per bed



ETL Process: From Raw File to Dimensional Model

The original dataset was provided as a single flat file containing all financial, operational, and location attributes, so several preprocessing steps were required to build the dimensional model.

- We first loaded the dataset into a staging table (STG_HOSPITAL_RAW) to support controlled transformation.
- We created the dimension tables and populated them using MERGE statements, ensuring that each business key (e.g., PROVIDER_CCN, ZIP_CODE, date) produced only one unique dimension record.
- For the location dimension, some ZIP codes had multiple CBSAs or counties. To enforce one row per ZIP/CITY/STATE, we selected consistent values using MIN() during the source query before the MERGE.
- After all dimensions were populated, we inserted data into FACT_FINANCIALS and FACT_OPERATIONS by joining the staging table with the appropriate dimension keys.
- This ETL process transformed the raw dataset into a clean, consistent star schema ready for analytic SQL and reporting.

Data Cubes Developed

Based on the dimensional model, two main analytical data cubes were developed to support the research questions.

Financial Performance Cube

Grain: One record per hospital per fiscal year
Source Fact: FACT_FINANCIALS Dimensions:

- Hospital
- Location
- Calendar

Purpose:
Supports analysis of profitability, financial structure, cost distribution, and differences across ownership type and geography.

Example Measures:
Total costs, total salaries, depreciation, inpatient charges, outpatient charges, combined charges, net income, wage related costs, cost to charge ratio.

Operational Capacity Cube

Grain: One record per hospital per fiscal year
Source Fact: FACT_OPERATIONS Dimensions:

- Hospital
- Location
- Calendar

Purpose:
Supports analysis of hospital size, staffing levels, bed capacity, bed day utilization, charity care, bad debt, and uncompensated care trends.

Example Measures:
Number of beds, bed days available, FTE employees, cost of charity care, bad debt expense, cost of uncompensated care.

Companion OLTP Design

HOSPITAL

Stores each hospital's identifying and structural information. (Same source as DIM_HOSPITAL, but not denormalized.)

COST_REPORT

Represents a single annual cost report submission. Connects the hospital to the fiscal year. Each hospital has multiple cost reports over time.

FINANCIAL_LINE_ITEM

Contains individual financial entries reported in the CMS cost report. Examples include salaries, depreciation, overhead, inpatient charges, outpatient charges. Each record represents one specific financial value for one cost report.

OPERATIONS_LINE_ITEM

Contains individual operational entries such as bed counts, bed days, staffing, charity care, bad debt, and uncompensated care. Each record represents one operational statistic for one cost report.

METRIC_REFERENCE

A lookup table that stores metric codes and labels used in the financial and operational line item tables. This supports a normalized design and avoids having hundreds of separate columns (as in the warehouse).

6. Exploratory Data Analysis

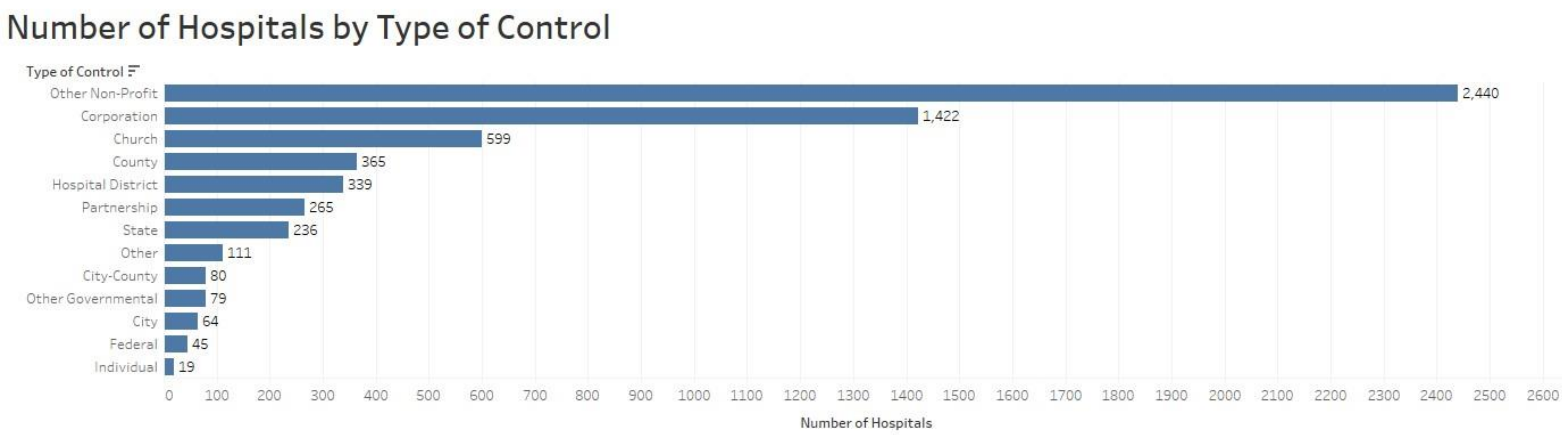
Grounded by the themes identified in the literature, we began by exploring the CMS dataset to confirm the presence of the same structural patterns observed in prior research. Before conducting the main analysis, we performed a brief exploratory review of the dataset to understand its structure and key characteristics. Using simple descriptive SQL queries, we examined the distribution of hospital ownership types, summarized major financial metrics, and evaluated basic efficiency indicators such as the cost-to-charge ratio. These preliminary insights helped confirm the dominant patterns in the data and guided the focus of our later analytic SQL queries.

Hospital Composition

```
SELECT
    type_of_control,
    COUNT(*) AS count_hospitals
FROM dim_hospital
GROUP BY type_of_control
ORDER BY count_hospitals DESC;
```

TYPE_OF_CON...	COUNT_HOSPITALS
2	2420
4	1397
1	598
9	363
11	339
5	255
10	233
6	108
8	80
13	79
12	64
7	45
3	18

Non-profit hospitals dominate the dataset (Types 2 and 1), followed by corporate/proprietary hospitals (Type 4). Government hospitals appear in much smaller counts across several categories. This confirms that most hospitals in the data operate under non-profit structures, which will influence profitability comparisons later in the analysis.



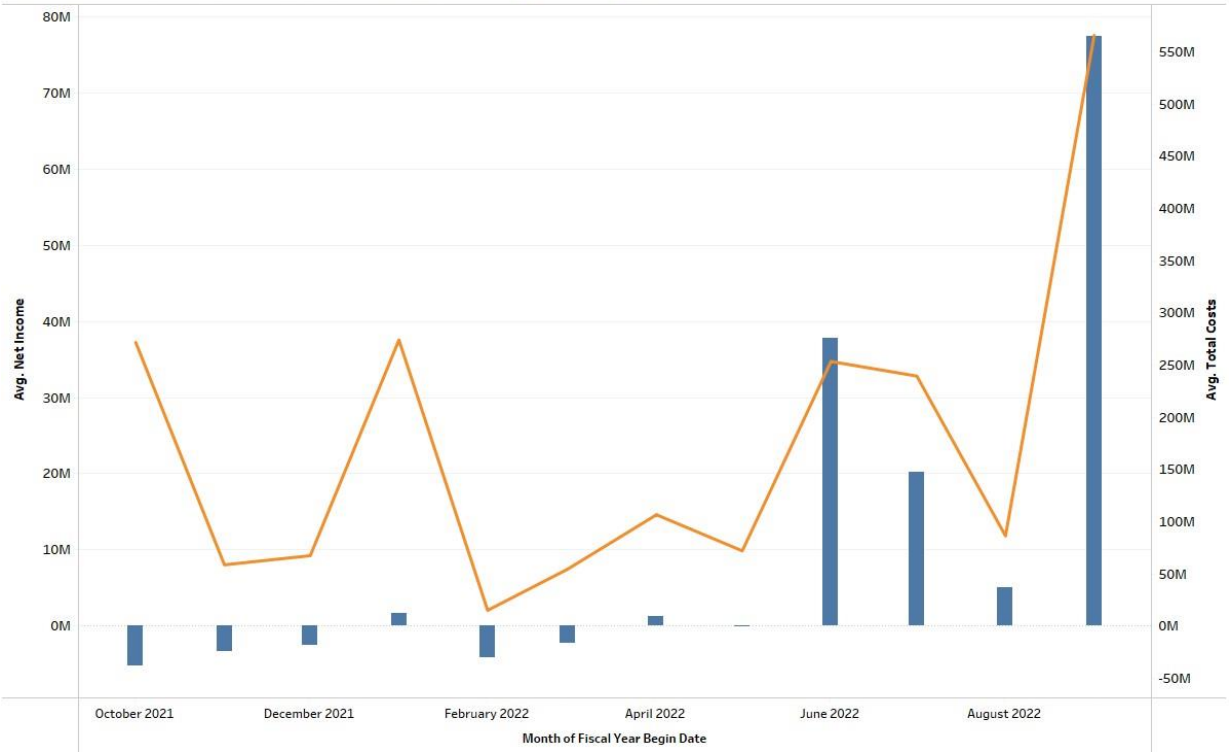
Key Financial Summary

```
SELECT
    ROUND(AVG(total_costs),2) AS TOTAL_COSTS,
    ROUND(AVG(net_patient_revenue),2) AS NET_PATIENT_REVENUE,
    ROUND(AVG(net_income),2) AS NET_INCOME
FROM fact_financials;
```

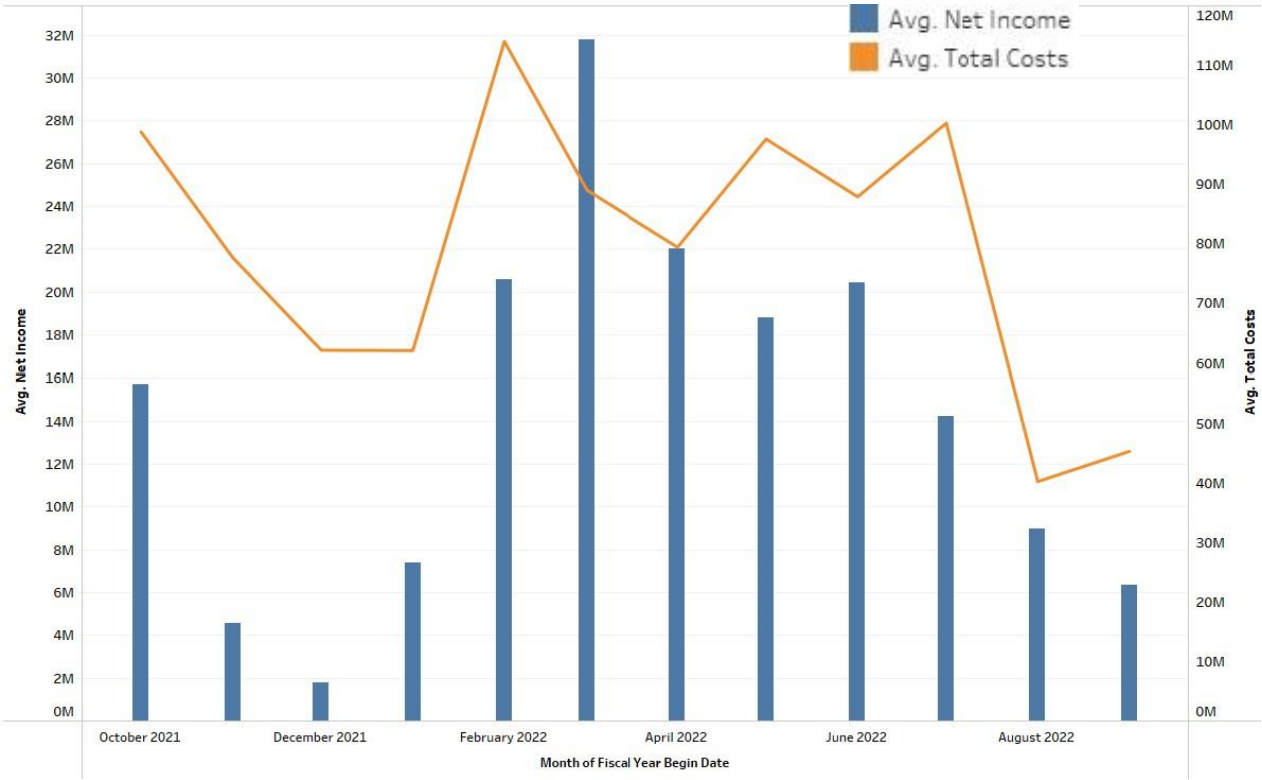
	TOTAL_COSTS	NET_PATIENT_REVENUE	NET_INCOME
1	183720781.84	222289183.21	7117900.53

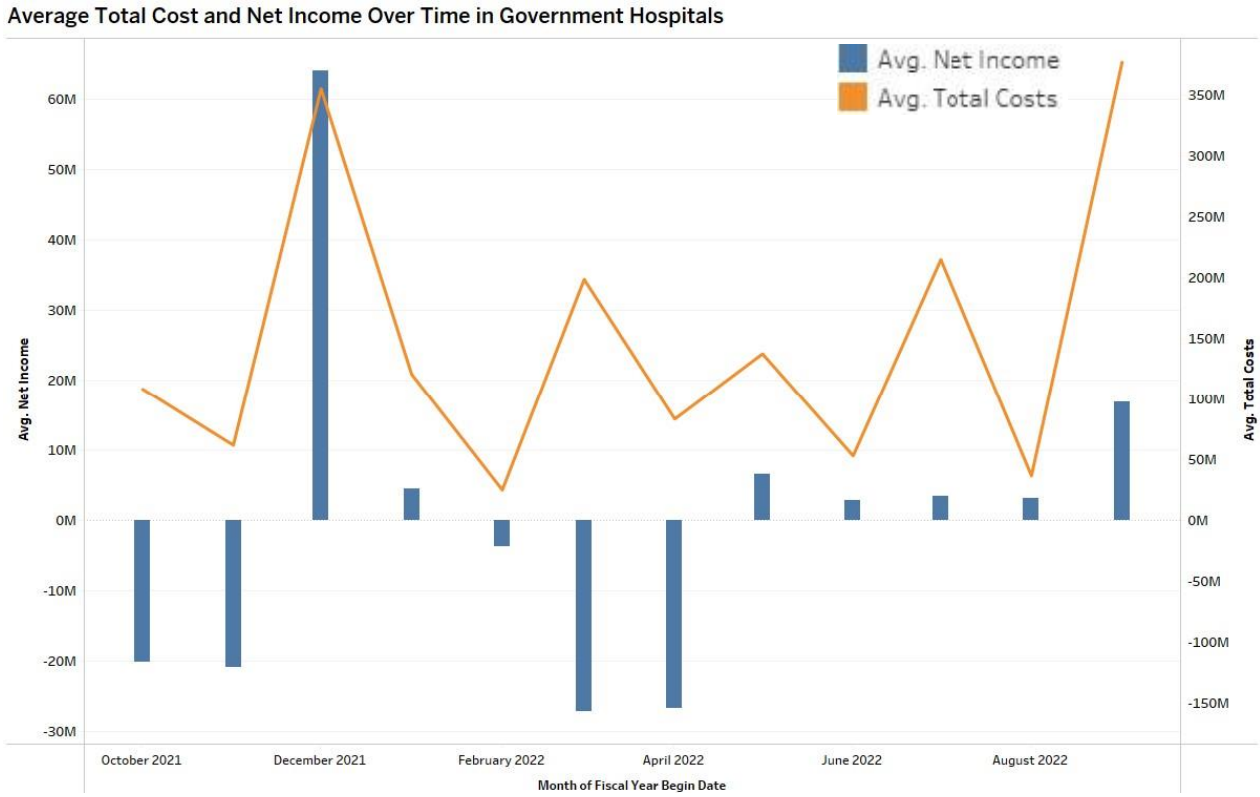
On average, hospitals operate with high costs and relatively low net income, indicating thin margins and motivating our deeper profitability analysis.

Average Total Cost and Net Income Over Time in Nonprofit Hospitals



Average Total Cost and Net Income Over Time in For-Profit Hospitals



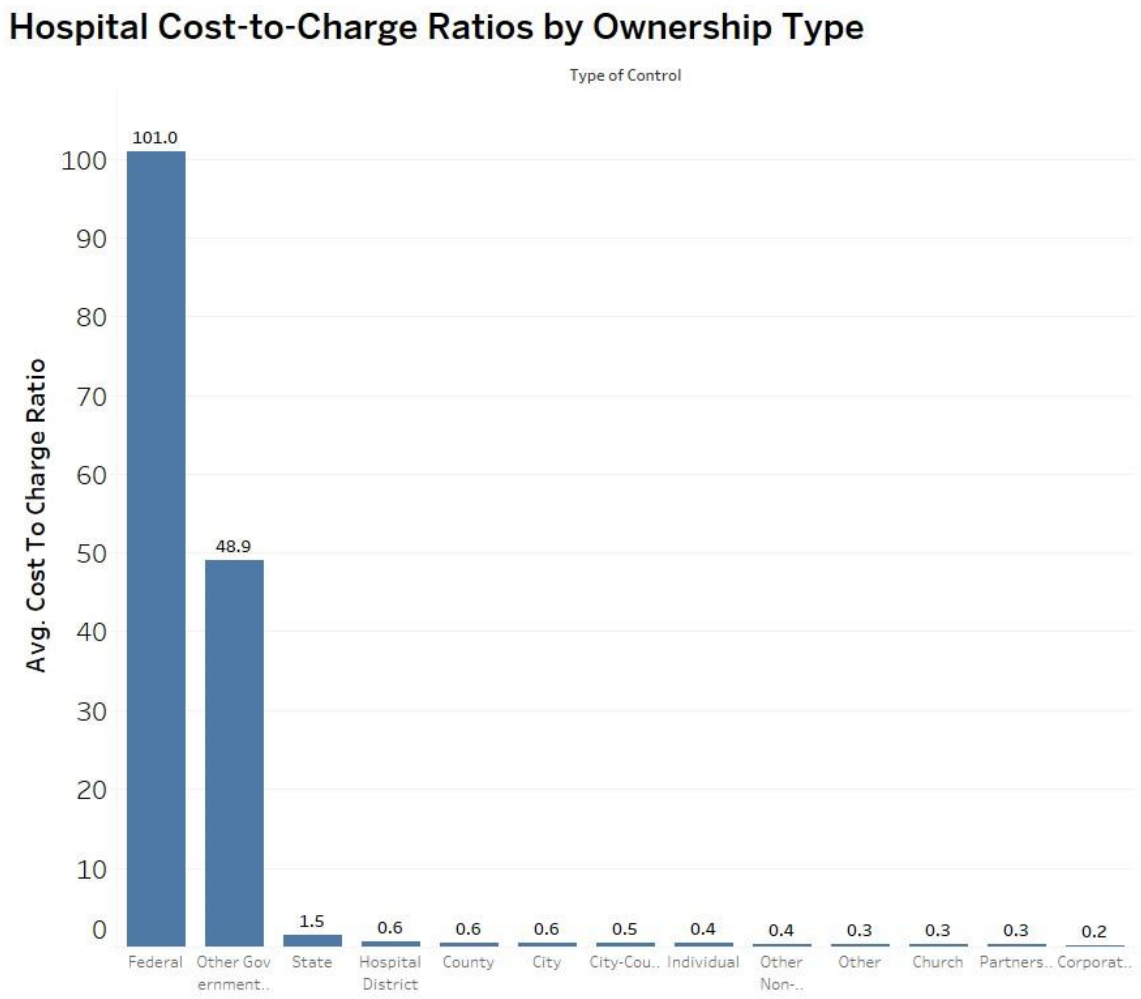


Efficiency/Cost Ratio Check

```
SELECT
    ROUND(AVG(cost_to_charge_ratio),4) AS AVG_COST_TO_CHARGE_RATIO FROM
fact_financials;
```

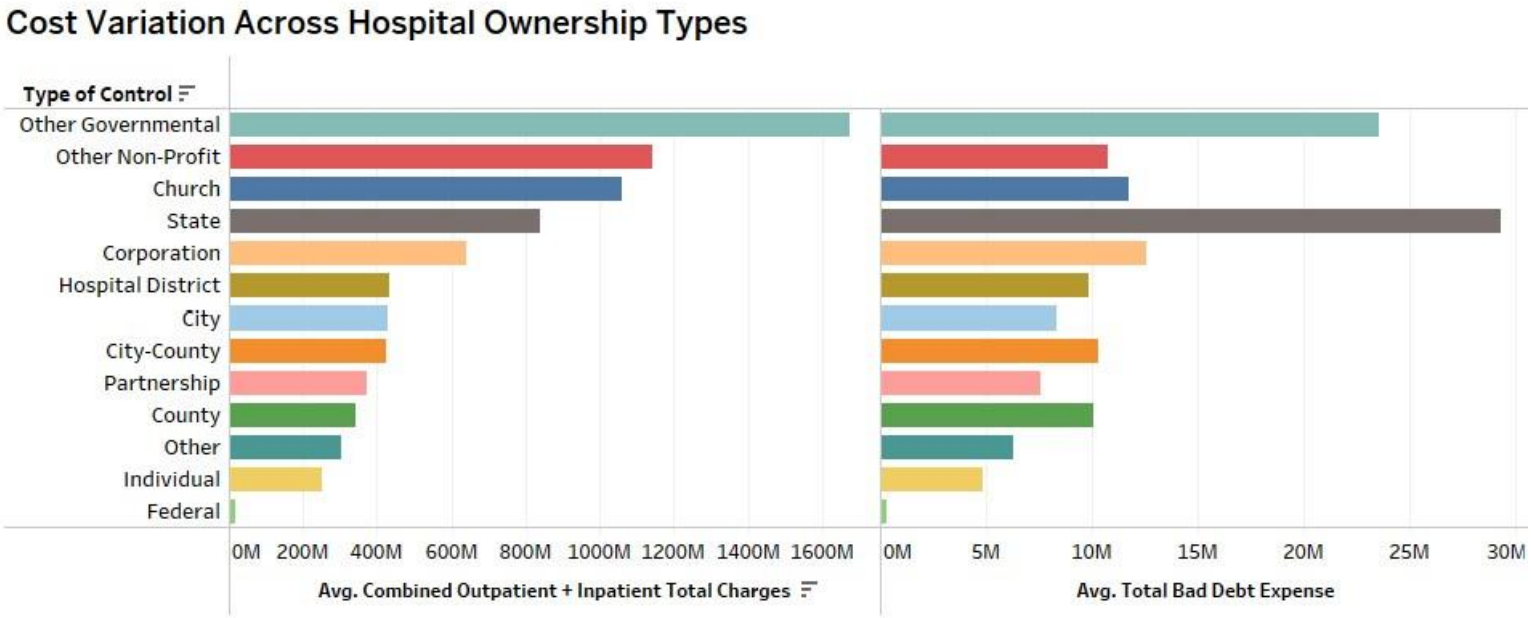
	AVG_COST_TO_CHARGE_RATIO
1	1.3498

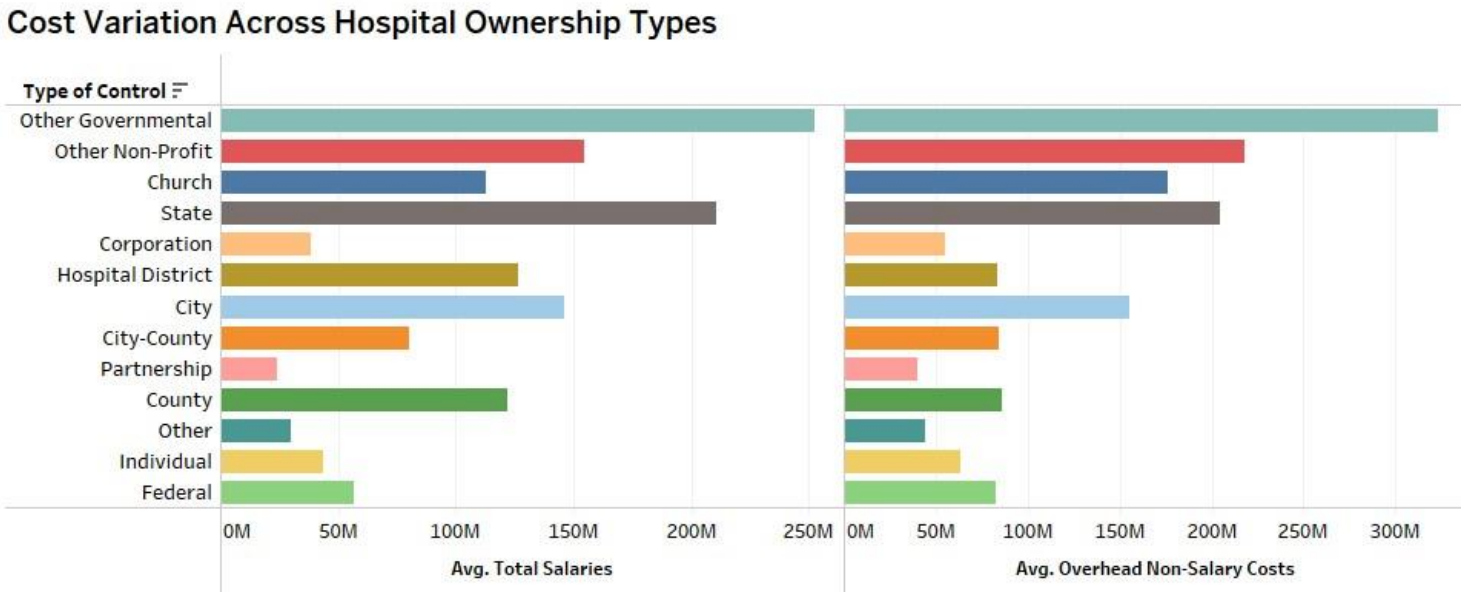
The average Cost-to-Charge Ratio is 1.3498, which means hospitals spend about 35% more than they charge. This shows that hospitals often operate with high costs and tight margins.



Costs by Hospital Ownership

By comparing metrics such as outpatient and inpatient total charges, bad debt expense, salaries, and overhead, we can identify which types of hospitals incur higher costs in specific areas, providing insight into how ownership may influence cost management and efficiency.

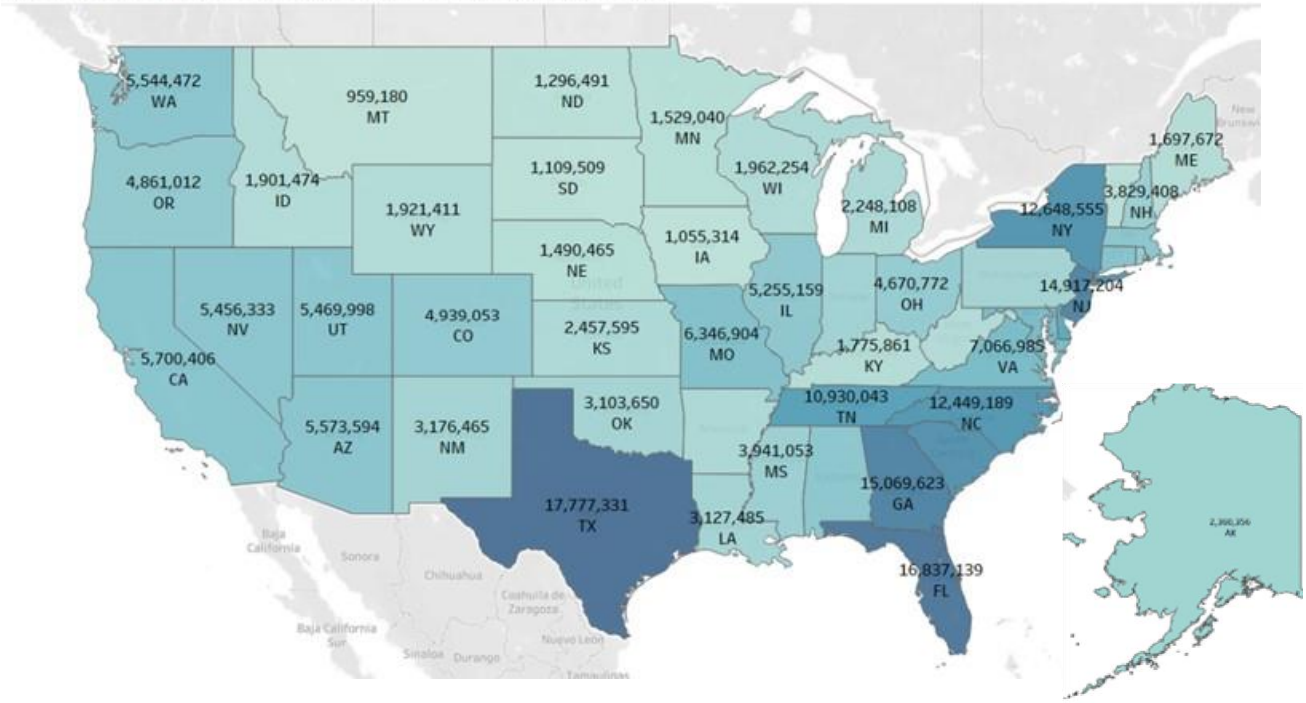




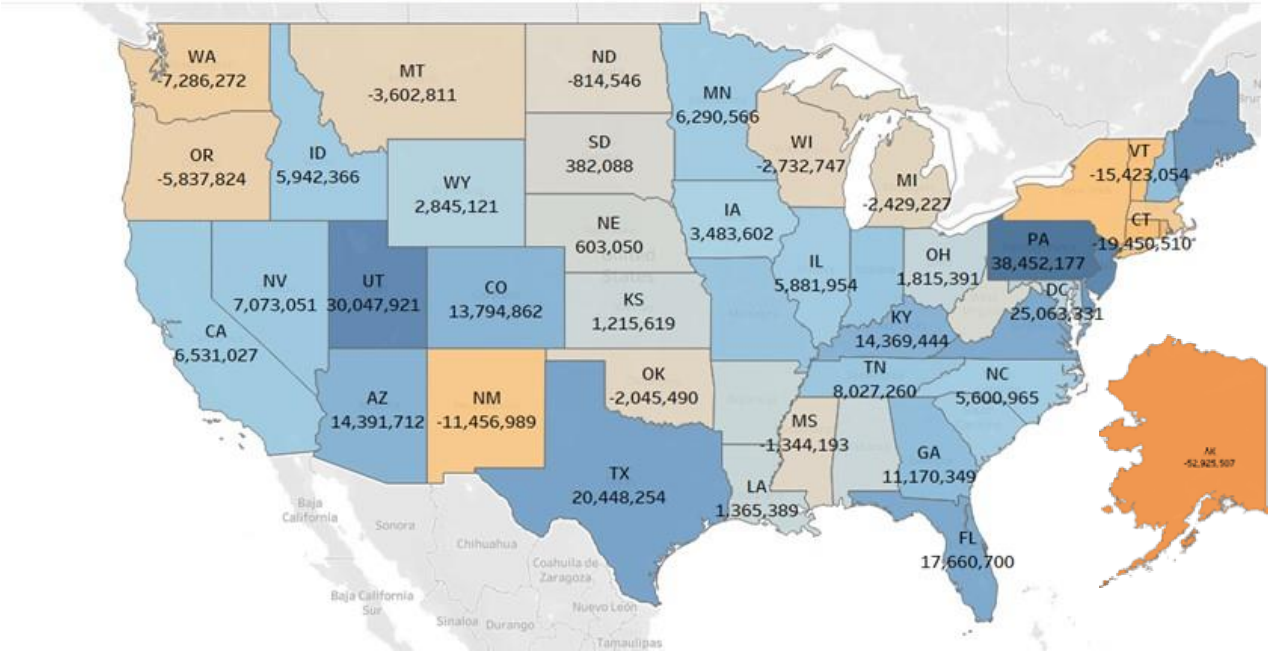
Hospital Charity Care and Net Income Across States

To complement the operational efficiency analysis, we also explore how financial conditions vary by state. By comparing average charity care spending and average net income, we can see which states' hospitals may be balancing uncompensated care obligations with tighter or more favorable margin.

Average Hospital Charity Care Spending by State



Average Hospital Net Income by State



These preliminary observations informed the design of the analytic SQL queries that follow.

7. Reporting, Modeling and Storytelling

As part of this project, the team developed the conceptual dimensional model, including separate dimension tables (Hospital, Location, Calendar) and fact tables (Financials, Operations), and implemented these tables for the analytical SQL work. The analytic SQL queries in this section were designed to directly address the project's two research questions concerning hospital profitability and operational efficiency. Each query draws on the star schema developed earlier, using ownership, location, financial metrics, and operational capacity as core dimensions. The strategy focuses on examining patterns identified in the literature review, such as differences by ownership type, geographic variation, labor intensity, and uncompensated care. Techniques including ROLLUP, ranking functions, NTILE classification, and lag comparisons were selected to highlight state-level variation, margin distribution across hospitals, quartile differences and year-over-year changes. Together, these queries form a structured analytical narrative that links the database design to meaningful business and policy insights.

SQL Analytic

Query 1: Profitability by Ownership and State (ROLLUP)

This query uses a ROLLUP across Type of Control and State to compare operating margins and average hospital size (beds and FTEs). It provides both detailed state-level values and higherlevel ownership totals.

```
SELECT
    CASE
        WHEN GROUPING(h.type_of_control) = 1 THEN 'ALL OWNERSHIP TYPES'
        ELSE h.type_of_control
    END AS type_of_control,

    CASE
        WHEN GROUPING(h.state_code) = 1 THEN 'ALL STATES'
        ELSE h.state_code
    END AS state_code,

    ROUND(
        (SUM(ff.net_patient_revenue) - SUM(ff.total_operating_expense))
        / NULLIF(SUM(ff.net_patient_revenue), 0),
        4
    ) AS operating_margin,

    ROUND(AVG(fo.number_of_beds), 2)           AS avg_beds,
ROUND(AVG(fo.fte_employees_on_payroll), 2) AS avg_fte

FROM fact_financials ff
JOIN fact_operations fo
    ON ff.rpt_rec_num = fo.rpt_rec_num
JOIN dim_hospital h
```

```
    ON ff.dim_hospital_id = h.dim_hospital_id
GROUP BY ROLLUP (h.type_of_control, h.state_code)
ORDER BY type_of_control, state_code;
```

	TYPE_OF_CONTROL	STATE_CODE	OPERATING_MARGIN	AVG_BEDS	AVG_FTE
1	1	AK	0.0286	119	766.91
2	1	AL	-0.0849	187.5	662.66
3	1	ALL STATES	-0.0561	166.78	951.71
4	1	AR	-0.0488	130	682.54
5	1	AZ	0.0635	197	1256.73
6	1	CA	-0.1404	185.59	1091.68
7	1	CO	-0.0851	318.5	1717.95
8	1	CT	-0.045	196.67	1184.54
9	1	FL	-0.1149	198.23	1085.73
10	1	GA	-0.1145	133	929.63
11	1	HI	-0.1392	160	800.14
12	1	IA	-0.0868	136.74	937.89
13	1	ID	-0.0296	129.5	1074.14
14	1	IL	0.0016	188.55	894.51
15	1	IN	0.1035	124.04	610.61
16	1	KS	-0.0904	116	612.66
17	1	KY	-0.0703	121	763.64
18	1	LA	-0.0028	220	1603.08
19	1	MA	-0.1457	85	474.58
20	1	MD	-0.1405	215.29	1772.86

Insight:

Operating margins are generally low or negative across most ownership types and states, confirming that hospitals operate with tight financial pressure. Larger hospitals (higher beds and FTEs) do not consistently show higher margins, suggesting that size alone does not guarantee better financial performance.

Query 2: Rank hospitals by margin within each state

This query computes each hospital's operating margin (when revenue > 0) and then uses RANK() OVER (PARTITION BY state_code) to rank hospitals within their state from highest to lowest margin.

```
WITH hospital_margin AS (
    SELECT
        h.state_code,
        h.hospital_name,
        h.type_of_control,
        -- margin only when revenue > 0
        CASE
            WHEN SUM(ff.net_patient_revenue) > 0 THEN
                (SUM(ff.net_patient_revenue) -
SUM(ff.total_operating_expense))
                / SUM(ff.net_patient_revenue)
```

```
        END AS operating_margin
    FROM fact_financials ff
    JOIN dim_hospital h
        ON ff.dim_hospital_id = h.dim_hospital_id
    GROUP BY
        h.state_code,
        h.hospital_name,
        h.type_of_control
) SELECT
state_code,
hospital_name,
type_of_control,
    ROUND(operating_margin, 4) AS operating_margin,
    RANK() OVER (
        PARTITION BY state_code
        ORDER BY operating_margin DESC
    ) AS state_margin_rank
FROM hospital_margin
WHERE operating_margin IS NOT NULL
ORDER BY state_code, state_margin_rank;
```

	STATE_CODE	HOSPITAL_NAME	TYPE_OF_CONTROL	OPERATING_MARGIN	STATE_MARGIN_RANK
1	AK	ALASKA REGIONAL HOSPITAL	4	0.4065	1
2	AK	MAT-SU REGIONAL MED CTR	4	0.2675	2
3	AK	PROVIDENCE SEWARD MEDICAL & CARE CEN	12	0.1752	3
4	AK	PROVIDENCE VALDEZ MEDICAL CENTER	12	0.0836	4
5	AK	PROVIDENCE ALASKA MEDICAL CENTER	1	0.0679	5
6	AK	CENTRAL PENINSULA GENERAL HOSPITAL	2	0.0555	6
7	AK	WRANGELL MEDICAL CENTER	7	0.0067	7
8	AK	PROV. KODIAK ISLAND MEDICAL CENTER	1	-0.0063	8
9	AK	BARTLETT REGIONAL HOSPITAL	8	-0.0081	9
10	AK	SOUTH PENINSULA HOSPITAL	12	-0.0343	10
11	AK	FAIRBANKS MEMORIAL	2	-0.0344	11
12	AK	CORDOVA COMMUNITY MEDICAL CENTER	12	-0.0547	12
13	AK	NORTH STAR HOSPITAL T18	4	-0.057	13
14	AK	PETERSBURG MEDICAL CENTER	12	-0.1065	14
15	AK	KETCHIKAN MEDICAL CENTER	1	-0.1308	15
16	AK	NORTON SOUND REGIONAL HOSPITAL	2	-0.2863	16
17	AK	ALASKA SPECIALTY HOSPITAL LLC	1	-0.2905	17
18	AK	SAMUEL SIMMONDS MEMORIAL HOSPITAL	7	-0.8167	18
19	AL	SOUTH BALDWIN REGIONAL MEDICAL CENTE	4	0.4642	1

Insight:

Within the same state, hospitals show big differences in profitability—some hospitals have strong positive margins while others are negative. This suggests that factors beyond location alone (e.g., management, cost structure, payer mix) drive financial performance, even among hospitals operating in the same environment.

Query 3: Quartiles of profitability (NTILE)

This query calculates each hospital's operating margin and then uses NTILE(4) to assign hospitals into four quartiles, from lowest (1) to highest (4) profitability.

```
SELECT
    h.hospital_name,
    h.state_code,
    h.type_of_control,
    ROUND(
        (SUM(ff.net_patient_revenue) - SUM(ff.total_operating_expense))
        / NULLIF(SUM(ff.net_patient_revenue), 0),
        4
    ) AS operating_margin,
    NTILE(4) OVER (
        ORDER BY
            (SUM(ff.net_patient_revenue) -
SUM(ff.total_operating_expense))
             / NULLIF(SUM(ff.net_patient_revenue), 0)
    ) AS margin_quartile
FROM fact_financials ff
JOIN dim_hospital h
    ON ff.dim_hospital_id = h.dim_hospital_id
GROUP BY
    h.hospital_name,
    h.state_code,
    h.type_of_control;
```

	HOSPITAL_NAME	STATE_CODE	TYPE_OF_CONTROL	OPERATING_MARGIN	MARGIN_QUARTILE
1	EAST CENTRAL REGIONAL HOSPITAL	GA	10	-158901702	1
2	EAST LA STATE MENTAL HEALTH SYSTEM	LA	10	-1465.7316	1
3	FLORIDA STATE HOSPITAL	FL	10	-72.0611	1
4	GREYSTONE PARK PSYCH. HOSPITAL	NJ	10	-36.6157	1
5	OSAWATOMIE STATE HOSPITAL	KS	10	-29.1312	1
6	TRENTON PSYCHIATRIC HOSPITAL	NJ	10	-24.014	1
7	SOUTH FLORIDA STATE HOSPITAL	FL	10	-21.1528	1
8	SOUTHERN VIRGINIA MENTAL HEALTH INST	VA	10	-18.195	1
9	WESTERN STATE HOSPITAL	VA	10	-16.8112	1
10	MILDRED MITCHELL-BATEMAN HOSPITAL	WV	10	-15.9614	1
11	WILLIAM R SHARPE JR HOSPITAL	WV	10	-15.4142	1
12	CO MENTAL HEALTH INSTITUTE - PUEBLO	CO	10	-14.4555	1
13	NORTHEAST FLORIDA STATE HOSPITAL	FL	10	-14.1717	1
14	ANCORA PSYCHIATRIC HOSPITAL	NJ	10	-12.9093	1
15	CHERRY HOSPITAL	NC	10	-12.3525	1
16	RICHMOND STATE HOSPITAL	IN	10	-11.4129	1
17	CENTER FOR FORENSIC PHYCHIATRY	MI	10	-10.4525	1
18	NORTH MISSISSIPPI STATE HOSPITAL	MS	10	-8.8512	1
19	MENDOTA MENTAL HEALTH INSTITUTE	WI	10	-7.9234	1

Insight:

Hospitals in the lowest quartile (1) have very large negative margins, and many of them are state-run mental health or psychiatric hospitals (Type 10). This shows that the worst financial performance is highly concentrated in specific government-owned facilities, rather than being spread evenly across all hospital types.

Query 4: Year-Over-Year Margin Change (LAG Window Function)

This query calculates each hospital's operating margin for each fiscal year and uses the LAG function to compute the year-over-year change. It allows us to see how hospital profitability shifts over time.

```
WITH hospital_year_margin AS (
    SELECT
        h.provider_ccn,
        h.hospital_name,
        c.year,
        CASE
            WHEN SUM(ff.net_patient_revenue) > 0 THEN
                (SUM(ff.net_patient_revenue) -
SUM(ff.total_operating_expense))
                / SUM(ff.net_patient_revenue)
        END AS operating_margin
    FROM fact_financials ff
    JOIN dim_hospital h
        ON ff.dim_hospital_id = h.dim_hospital_id
    JOIN dim_calendar c
        ON ff.dim_calendar_id = c.dim_calendar_id
    GROUP BY
        h.provider_ccn,
        h.hospital_name,
        c.year ),
margin_with_lag AS (
SELECT
provider_ccn,
hospital_name,
year,
        operating_margin,
        LAG(operating_margin) OVER (
PARTITION BY provider_ccn
            ORDER BY year
        ) AS prev_year_margin
FROM hospital_year_margin
WHERE operating_margin IS NOT NULL
```

```
) SELECT
provider_ccn,
hospital_name,
year,
    ROUND(operating_margin, 4) AS operating_margin,
    ROUND(prev_year_margin, 4) AS prev_year_margin,
ROUND(
        operating_margin - prev_year_margin,
        4
    ) AS yoy_change
FROM margin_with_lag
WHERE prev_year_margin IS NOT NULL
ORDER BY hospital_name, year;
```

	PROVIDER_CCN	HOSPITAL_NAME	YEAR	OPERATING_MARGIN	PREV_YEAR_MARGIN	YOY_CHANGE
1	390333	AHN HEMPFIELD NEIGHBORHOOD HOSPITAL	2023	-0.0037	-0.0174	0.0137
2	390334	AHN WEXFORD HOSPITAL	2023	0.0554	-0.0869	0.1423
3	340123	AMERICAN HEALTHCARE SYSTEMS	2022	0.097	-0.2143	0.3113
4	441303	ASCENSION ST THOMAS THREE RIVERS	2023	-0.0892	-1.2858	1.1967
5	171382	ATCHISON HOSPITAL ASSOCIATION	2022	-0.1174	-0.0451	-0.0724
6	234045	BEAUMONT BEHAVIORAL HEALTH	2023	-0.0852	-1.5504	1.4652
7	340075	BLUE RIDGE HEALTHCARE HOSPITALS	2023	-0.0113	-0.053	0.0418
8	340131	CAROLINAEAST MEDICAL CENTER	2023	-0.0694	-0.0293	-0.04
9	170175	CENTURA ST CATHERINE - DODGE CITY	2023	0.1142	0.0681	0.0462
10	60044	CENTURA ST. ELIZABETH HOSPITAL	2023	-0.0059	-0.0444	0.0385
11	43031	CHI ST. VINCENT SHERWOOD REHABILITAT	2023	0.2075	0.1799	0.0276
12	63037	COBALT REHAB HOSP WESTMINSTER	2022	-0.3656	-2.9487	2.583
13	490019	CULPEPER MEMORIAL HOSPITAL	2023	0.0933	0.0176	0.0757
14	153025	ENCOMPASS HEALTH DEACONESS REHABILIT	2023	0.2971	0.3113	-0.0141
15	403025	ENCOMPASS HEALTH REHABILITATION HOSP	2023	0.1474	0.2505	-0.1031
16	453029	ENCOMPASS HEALTH REHABILITATION HOSP	2023	0.0604	0.0642	-0.0038
17	453047	ENCOMPASS HEALTH REHABILITATION HOSP	2023	0.1615	0.1754	-0.0139
18	673034	ENCOMPASS HEALTH REHABILITATION HOSP	2023	0.2287	0.1592	0.0695
19	13030	ENCOMPASS HEALTH REHABILITATION HOSP	2023	0.211	0.3212	-0.1103
20	33029	ENCOMPASS HEALTH REHABILITATION HOSP	2023	-0.0076	0.3847	-0.3923
21	183027	ENCOMPASS HEALTH REHABILITATION HOSP	2023	0.1036	0.0761	0.0275

Insight:

Most hospitals show small year-to-year changes in margin, indicating stable financial performance despite tight margins. Large positive or negative swings are uncommon and usually concentrated in specific hospitals, suggesting that major financial shifts tend to be hospitalspecific rather than industry-wide.

8. Conclusion

Limitations

Although this project provides a structured view of hospital financial and operational performance, several limitations should be acknowledged. First, the CMS Hospital Cost Report is self-reported data, and hospitals vary in their accounting practices, reporting completeness, and interpretation of CMS definitions. These inconsistencies can affect comparability across facilities and states. Second, the dataset reflects annual aggregated values rather than patient-level or service-line granularity, which limits the ability to analyze variation within each hospital or across specific types of care. Third, operational fields such as FTE counts and bed-day utilization are not uniformly reported, which reduces sample size for some queries and may bias results toward hospitals with stronger administrative capacity. Finally, the dimensional model used in this project simplifies the operational environment to support analytical clarity. This design does not capture more complex relationships that would be present in a full OLTP system.

Findings

This project demonstrates how a well-designed dimensional data warehouse can turn complex Medicare Cost Report data into meaningful insights about hospital financial performance, operational capacity, and the pressures created by uncompensated care. By separating financial, operational, geographic, and temporal attributes into a star schema, the model enables flexible analysis across ownership types, states, counties, and fiscal years, mirroring the analytical approaches used in contemporary health services research.

The analysis shows clear and consistent patterns. First, ownership type remains one of the strongest predictors of financial outcomes. For-profit hospitals generally maintain higher operating margins. Nonprofit hospitals operate with thinner surpluses and government hospitals, especially state-run psychiatric facilities, show the greatest financial strain. These findings closely parallel recent academic studies, reinforcing that structural differences, not just managerial choices, shape margin performance nationwide.

Labor costs continue to be a dominant driver of financial stability. Rising wages and the postpandemic shift toward contract nursing elevate operational expenses across all ownership types, squeezing margins even in hospitals with strong revenue performance. Operational metrics reveal a similar pattern. Larger hospitals with substantial bed capacity often achieve better cost efficiency, but size alone does not guarantee profitability, especially in teaching or specialty hospitals carrying high fixed staffing costs.

Finally, uncompensated care emerges as one of the most persistent burdens on the system. Charity care and bad debt remain concentrated in government and safety-net hospitals, contributing to weaker cost-to-charge ratios and negative margins. Geographic context further amplifies these disparities. Rural hospitals continue to face lower volumes and fewer commercial payers, while urban hospitals benefit from larger markets but operate in higher-cost environments.

Taken together, these results underscore the value of integrating financial, operational, and geographic data in a single analytical environment. The dimensional model not only supports required SQL analyses, such as window functions, ranking, rollups, and derived efficiency ratios, but also provides a platform for future work, including OLAP cubes, time-series dashboards, and predictive modeling of hospital performance.
Ultimately, the project highlights that data-driven insights are essential for hospitals, administrators, and policymakers seeking to navigate the financial pressures of modern healthcare. By using structured data and reproducible analytics, stakeholders can better identify where resources are strained, which hospitals face the greatest operational challenges, and how financial patterns differ across the healthcare system. This approach supports more informed decision-making, more equitable allocation of resources, and a deeper understanding of the structural factors that shape hospital performance in the United States.

Future Work

There are several directions for expanding and strengthening this analysis. A natural next step is to incorporate additional reporting periods to create a richer longitudinal dataset that allows more precise evaluation of year-over-year trends. The model could also be enhanced with more detailed geographic information, such as Health Service Areas or Regional Health Market indicators, to improve regional comparisons. Another promising direction is to integrate external datasets, such as Medicare case mix indexes, community socioeconomic indicators, or staffing market analyses, to explain variation beyond what is captured in cost reports alone. On the modeling side, future work could include predictive analytics that estimate expected margins based on operational and financial inputs or cluster analysis to identify common profiles among hospitals. Finally, expanding the OLTP component into a more complete representation of transaction-level activity would support deeper analysis of resource flows within hospitals.

9. References

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